

Finite-Time Gradient Method Based Adaptive Model-Free Predictive Current Control With Harmonic Suppression

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Abstract—To improve the control performance of permanent magnet synchronous motor (PMSM) drive systems, this article proposes a model-free predictive current control method based on adaptive extended state observer (ESO) and considering harmonic suppression strategy. First, the article analyzes the controller gain of traditional ESO and concludes that controller gain mismatch can lead to system instability. Based on this, an adaptive ESO is proposed, which estimates the controller gain adaptively using the finite-time gradient method (FGM) to optimize the control performance degradation caused by controller gain mismatch in traditional ESO. Then, a current harmonic controller was proposed to suppress a series of low-order harmonics, further improving the steady-state performance of the control system. Finally, the experimental results prove the overall performance and parameter robustness of the method.

Index Terms—Adaptive extended state observer (AESO), current harmonic suppression, model-free predictive current control (MFPC), permanent magnet synchronous motor (PMSM), robustness.

I. INTRODUCTION

PERMANENT magnet synchronous motors (PMSMs) are widely used in high-precision applications such as computer numerical control machine tools, robotics, and aerospace due to their simple structure, high-power density, and high efficiency [1], [2]. Despite these advantages, a PMSM is a complex control object with nonlinearity, strong coupling, and time-varying parameters. Thus, appropriate control strategies are needed to achieve good control performance [3], [4], [5].

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There are mainly two mature methods in terms of PMSM control: direct torque control (DTC) and field-oriented control (FOC) [6], [7]. FOC offers high steady-state performance but requires tedious PI tuning, while DTC provides excellent dynamics but suffers from large torque ripple [8], [9].

The model predictive control (MPC) method has recently been widely used in PMSMs due to its simple control structure, fast dynamic response, and good steady-state performance [10], [11], [12]. The main control idea of MPC is to observe the error between the predictive model output and the actual quantity of the control system to control feedback correction, perform online optimization, and calculate the optimal control parameters to achieve the desired output effect. Based on the distinct control objectives, MPC can be divided into two approaches, which are model predictive current control (MPCC) and model predictive torque control (MPTC). In MPTC, the voltage vector with the smallest difference between reference and actual values of torque and magnetic flux is selected as the optimal voltage vector [13], [14]. Compared with the MPTC method, MPCC uses a current prediction model and a cost function to select the optimal voltage vector. The MPCC method is simpler because it avoids weighting factor design [15], [16].

However, as a model-based algorithm, MPC heavily relies on the accuracy of machine parameters [17]. During motor operation, machine parameters may vary with changes in operating conditions; furthermore, the measured motor parameters may also have measurement errors, which will adversely affect the performance of the motor system. Therefore, to improve the robustness of MPCC to parameter changes, researchers have developed different methods, including parameter identification methods, disturbance observer-based methods, and model-free control methods [18], [19], [20].

In recent years, parameter identification methods have become an important means to solve the problems of parameter mismatch [21]. In [22], an online parameter identification technique based on recursive least squares (RLSs) was applied to identify resistance values. In [23], an improved deadbeat predictive current control (DPCC) method was proposed, accurately identifying inductance and magnetic flux parameters based on the extended Kalman filter (EKF). However, parameter identification often requires

complex algorithm design, which undoubtedly increases computation time.

Another method is to introduce a disturbance observer in the system, which can achieve robust control by observing and compensating for disturbances caused by parameter changes. In [24], a robust predictive current control method based on the Luenberger observer was proposed. However, the Luenberger observer is mainly suitable for linear systems and has insufficient observation accuracy for highly nonlinear motor models. In [25], an improved DPCC scheme based on sliding mode was proposed, and a high order sliding mode observer (SMO) was designed to estimate and compensate for disturbances in the speed and current loops. However, the inherent chattering phenomenon in sliding mode controllers can lead to system instability and performance degradation. In contrast, extended state observer (ESO) has a simple structure and exhibits strong parameter robustness, making it more suitable for observing unknown disturbances compared to Luenberger observer and SMO. In [26], an ESO based on the self-stable region approach and the Lyapunov theory was designed, effectively improving the robustness of the finite-set MPTC method.

Nowadays, researchers have proposed model-free predictive control methods, which use the input and output of the system to establish mathematical models instead of physical models to describe the system state of the motor, reducing the dependence on motor parameters [27], [28]. Zhang et al. [29] proposed an improved model-free predictive current control (MFPC) method, which employs a first-order ultralocal model for the PMSM drive system and predicts future current values by utilizing current and past current information measured through ESO. This method does not require motor parameters and is robust. However, in the article, the controller gain in ESO was set to the reciprocal of the initial inductance, which goes against the original intention of studying MFPC. Furthermore, the parameter tuning issue presents another challenge within the field of model free control. Addressing this, Wang et al. proposed a prior-optimized model-free adaptive control (POM-FACC) method [30]. The approach systematically tackles the tuning problem by employing a multiobjective genetic algorithm to automatically preoptimize the key internal parameters of the controller. This ensures high performance and robustness without relying on motor parameters or manual adjustments, thereby more fully realizing the potential of model-free methods.

Traditional MFPC primarily focuses on compensating for disturbances caused by parameter mismatches and model uncertainties. However, its capability to suppress periodic harmonic disturbances is limited. In practice, due to various factors, there are a large number of harmonics in the three-phase current of the motor. In addition to the high-frequency current harmonics induced by pulse width modulation (PWM), there are various other sources that can cause current harmonics, including motor design defects, inverter nonlinearity, and other factors [31]. The presence of these harmonics will lead to motor torque fluctuations, vibration, and noise, which will affect the control performance of the system. In recent years, various methods have been studied to suppress current harmonics. In [32], an

adaptive notch filter (ANF) was applied to a shunt active power filter system to improve the dynamic response ability of harmonic detection. Dai et al. [33] proposed a current harmonic suppression method based on adaptive harmonic reference current injection to compensate for the effects of voltage source inverter (VSI) nonlinearity and back electromotive force (EMF) harmonics in PMSMs. Qiu et al. [34] proposed a dead-time compensation method for PMSMs based on adaptive linear neurons (ADALINE). However, Dai et al. [33] use two estimators and two compensators to generate compensation voltages for the dq axis, while the method in [34] requires real-time calculation and adjustment of the injected harmonic current during operation, making the control algorithms in both articles quite complex.

In this article, an MFPC method with adaptive extended state observer (AESO) and current harmonics suppression strategy is proposed. The contributions of this article are as follows.

- 1) The traditional ultralocal mathematical model of PMSM is improved to adapt to the design of AESO, enabling us to calculate the controller gain α through F_d estimated by the AESO. Meanwhile, the controller gain is updated in real-time via the finite-time gradient method (FGM), and stability analysis is conducted, which effectively addresses the issue of degraded control performance in conventional ESOs caused by mismatched controller gains.
- 2) A harmonic current control method is proposed to address the issue of harmonics in the three-phase current of PMSMs. The method effectively suppresses low-order harmonic currents, further improving the performance of the control system.

The rest of this article is organized as follows. Section II introduces a conventional MFPC-ESO method based on an ultralocal model. Section III presents an improved MFPC-AESO scheme. Section IV proposes the harmonic current controller. Finally, Section V presents the relevant experimental results to verify the effectiveness of the proposed scheme, followed by the conclusion in Section VI.

II. CONVENTIONAL MFPC WITH ESO

A. Conventional MFPC

Due to the same dq inductance in surface-mounted permanent magnet synchronous motor (SPMSM), the stator voltage can be described as

$$\begin{cases} u_d = R_s i_d + L_s \frac{di_d}{dt} - \omega_e L_s i_q \\ u_q = R_s i_q + L_s \frac{di_q}{dt} + \omega_e (L_s i_d + \psi_f) \end{cases} \quad (1)$$

where u_d and u_q are the d - and q -axis voltage, i_d and i_q are the d - and q -axis currents, and L_s is the d - and q -axis inductances, respectively. R_s is the stator resistance, ψ_f is the permanent magnetic flux, and ω_e is the rotor electrical speed.

The torque generated by the PMSM can be given by

$$T_e = \frac{3}{2} n_p i_q \psi_f \quad (2)$$

where T_e is the torque and n_p is the pole pair of the motor.

By using the first-order Euler, (1) can be discretized as

$$i_{dq}(k+1) = A i_{dq}(k) + B u_{dq}(k) + C(k) \quad (3)$$

where $i_{dq}(k+1) = \begin{bmatrix} i_d(k+1) \\ i_q(k+1) \end{bmatrix}$, $i_{dq}(k) = \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix}$, $u_{dq}(k) = \begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix}$, $A = \begin{bmatrix} 1 - \frac{R_s T_s}{L_s} & T_s \omega_e(k) \\ -T_s \omega_e(k) & 1 - \frac{R_s T_s}{L_s} \end{bmatrix}$, $B = \begin{bmatrix} \frac{T_s}{L_s} & 0 \\ 0 & \frac{T_s}{L_s} \end{bmatrix}$, $C = \begin{bmatrix} 0 \\ -\frac{\psi_f}{L_s} T_s \omega_e(k) \end{bmatrix}$.

B. ESO Based on Ultralocal Model

The first-order ultralocal model is expressed as

$$\frac{dy}{dt} = \alpha u + F \quad (4)$$

where u and y are the control variable and output variable, respectively, α represents a scaling factor, and F represents the unknown disturbances of the system.

According to (4), the ultralocal mathematical model of SPMSM can be described as

$$\begin{cases} \frac{di_d}{dt} = \frac{1}{L_s}(u_d - R_s i_d + \omega_e L_s i_q) \\ \frac{di_q}{dt} = \frac{1}{L_s}(u_q - R_s i_q - \omega_e L_s i_d - \omega_e \psi_f). \end{cases} \quad (5)$$

Based on (4) and (5), the discrete-time form of the predictive PMSM model can be expressed as

$$i_{dq}(k+1) = i_{dq}(k) + \alpha T_s u_{dq}(k) + T_s F_{dq}(k) \quad (6)$$

where $\alpha = 1/L_s$, $F_d = 1/L_s(-R_s i_d + \omega_e L_s i_q)$, $F_q = 1/L_s(-R_s i_q - \omega_e L_s i_d - \omega_e \psi_f)$, and T_s represents the sampling frequency.

The ESO can be constructed as follows:

$$\begin{cases} e_{rr} = \hat{i}_{dq} - i_{dq} \\ \frac{d\hat{i}_{dq}}{dt} = \hat{F}_{dq} + \alpha_0 u_{dq} + \beta_1 e_{rr} \\ \frac{d\hat{F}_{dq}}{dt} = \beta_2 e_{rr} \end{cases} \quad (7)$$

where $\hat{i}_{dq}$ and $\hat{F}_{dq}$ are the estimation of i_{dq} and F_{dq} , α_0 is the gain of the controller input, and β_1 and β_2 are the error feedback gain coefficients of ESO.

To ensure the stability and desired dynamic performance of the conventional ESO, its parameters are designed based on the pole placement method, as rigorously proven in [29]. The characteristic equation of the observer is configured such that its poles are placed within the unit circle in the z -domain, which is a sufficient condition for stability. In this work, appropriate error feedback gain coefficients β_1 and β_2 can be selected to ensure the stability of the observer

$$\begin{cases} \beta_1 = -2\omega_b \\ \beta_2 = -\omega_b^2 \end{cases} \quad (8)$$

where ω_b is the bandwidth of ESO, and the design of its value will determine the dynamic performance and stability of the current loop. By selecting appropriate values of ω_b , the conventional ESO can achieve good control performance.

Using the forward Euler discretization method, (7) can be discretized into

$$\begin{cases} e_{rr}(k) = \hat{i}_{dq}(k) - i_{dq}(k) \\ \hat{i}_{dq}(k+1) = \hat{i}_{dq}(k) + T_s(\hat{F}_{dq}(k) + \alpha_0 u_{dq}(k) + \beta_1 e_{rr}(k)) \\ \hat{F}_{dq}(k+1) = \hat{F}_{dq}(k) + T_s \beta_2 e_{rr}(k). \end{cases} \quad (9)$$

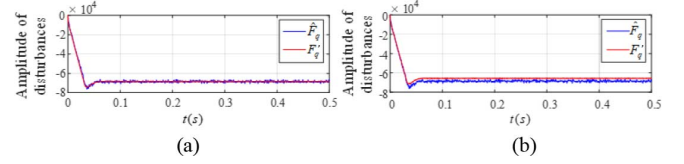


Fig. 1. Simulation about $\hat{F}_{dq}$ and F'_{dq} . (a) $\Delta\alpha = 0$. (b) $\Delta\alpha = 50$.

According to (6), u_s^k can be described as

$$u_{dq}(k) = \frac{i_{dq}(k+1) - i_{dq}(k)}{\alpha_0 T_s} - \frac{\hat{F}_{dq}(k)}{\alpha_0} \quad (10)$$

where F is replaced by $\hat{F}$ because F cannot be directly obtained.

Compared with traditional DPCC based on the accurate model, MFPPCC no longer requires stator resistance and flux parameters of the motor. However, the design of controller gain still depends on the initial inductance parameters, which is inconsistent with the original intention of MFPPCC.

III. PROPOSED MFPPCC WITH AESO

A. Analysis of ESO Under Controller Gain Mismatch

Based on (4) and (5), the continuous time form of the predictive PMSM model can be expressed as

$$\begin{cases} \frac{di_d}{dt} = \alpha u_d + F_d \\ \frac{di_q}{dt} = \alpha u_q + F_q. \end{cases} \quad (11)$$

Now assuming that the controller gain is inaccurate, the ultralocal mathematical model of PMSM can be described as

$$\begin{cases} \frac{di_d}{dt} = \alpha_a u_d + F'_d \\ \frac{di_q}{dt} = \alpha_a u_q + F'_q \end{cases} \quad (12)$$

where α_a is the gain of the controller input, and its value will be adaptively adjusted in the subsequent AESO design.

Then, the theoretically disturbance value can be expressed as follows:

$$\begin{cases} F'_d = \Delta\alpha u_d - \frac{R_s}{L_s} i_d + \omega_e i_q \\ F'_q = \Delta\alpha u_q - \frac{R_s}{L_s} i_q - \omega_e i_d - \frac{\psi_f}{L_s} \omega_e \end{cases} \quad (13)$$

where $\Delta\alpha = \alpha - \alpha_a$ is the difference between the initial inductance parameter and the actual inductance parameter.

It can be found in (13) that the first term of F'_{dq} is caused by the inductance mismatch and will introduce high frequency component into the total disturbance. In Fig. 1, as the value of $\Delta\alpha$ on the q -axis increases, there is a deviation between the estimated disturbance and the actual disturbance. Therefore, inaccurate controller gain results in a larger deviation between the theoretical disturbance F'_{dq} and the estimated disturbance $\hat{F}_{dq}$, which degrades the control performance.

If the actual controller gain is inconsistent with the reciprocal of the initial inductance, it will directly deteriorate the control performance of the conventional ESO. Thus, an AESO method is introduced to eliminate the negative impact caused by controller gain mismatch.

B. Design of AESO

Prior to designing the AESO, it is necessary to clarify the composition of the disturbance term F_{dq} in the ultralocal model, particularly the nonzero characteristics of F_d caused by practical system nonidealities. In PMSM drive systems, inverter deadtime and the nonlinear characteristics of power devices result in voltage distortion in the d - q axis, further inducing nonzero disturbance components in F_d and F_q . Even under the $i_d = 0$ control strategy, the voltage distortion caused by deadtime cannot be overlooked. To compensate for the disturbances induced by dead time and ensure the accuracy of subsequent disturbance observation, this article introduces a harmonic suppression method based on an adaptive neural network framework, which will be elaborated in the next section. After applying this method, the disturbance term F_{dq} in the ultralocal model is primarily determined by inductance mismatch and system dynamic errors, thereby laying the foundation for the subsequent design of the AESO.

To address the impact of controller gain mismatch, the terms $\omega_e i_q$ and $\omega_e i_d$ containing system state variables in F_{dq} can be separated, resulting in an improved ultralocal model of SPMSM, whose expression is as follows:

$$\begin{cases} \frac{di_d}{dt} = \alpha_a u_d + \omega_e i_q + F'_d \\ \frac{di_q}{dt} = \alpha_a u_q - \omega_e i_d + F'_q \end{cases} \quad (14)$$

Compared (12) and (14), the theoretical disturbance value can be rewritten as

$$\begin{cases} F'_d = \Delta \alpha u_d - \frac{R_s}{L_s} i_d \\ F'_q = \Delta \alpha u_q - \frac{R_s}{L_s} i_q - \frac{\psi_f}{L_s} \omega_e \end{cases} \quad (15)$$

The AESO can be designed as follows:

$$\begin{cases} e_{rr} = \hat{i}_{dq} - i_{dq} \\ \frac{d\hat{i}_{dq}}{dt} = D i_{dq} + \hat{F}_{dq} + \alpha_a u_{dq} + \beta_1 e_{rr} \\ \frac{d\hat{F}_{dq}}{dt} = \beta_2 e_{rr} \end{cases} \quad (16)$$

where $D = \begin{bmatrix} 0 & \omega_e \\ -\omega_e & 0 \end{bmatrix}$.

Under the control of $i_d = 0$, according to (15), the observed value of the disturbance can be expressed as

$$F'_d \approx \Delta \alpha u_d. \quad (17)$$

As previously mentioned, the value of α_a is always selected as $1/L_s$, the reciprocal of inductance. However, (17) indicates that when the controller gain is mismatched, the disturbance value F'_d will significantly change. To improve the robustness of inductance parameters, an α_a adaption method based on FGM is proposed.

The criterion for FGM adopted can be expressed as

$$\begin{cases} \dot{s} = -\theta \text{sgn}(\nabla f(s)) \\ \dot{\theta} = -\lambda \theta + \rho \|\nabla f(s)\| \end{cases} \quad (18)$$

where $f(s)$ is a function with an extreme point s_0 , θ stands for the iterative step size, and λ and ρ are two parameters to adjust θ . A larger ρ results in a smaller step size θ , making the algorithm less sensitive to gradient noise and parameter

variations, thereby enhancing stability, but potentially at the cost of slower convergence. A larger λ leads to a larger step size θ , accelerating convergence but increasing the risk of oscillations and degrading system stability. These two parameters must be codesigned. In addition, it has been proved in [34] that (18) can converge to the extreme point within a finite time by selecting appropriate parameters. The convergence condition can be expressed as follows:

$$t \leq \frac{2\pi}{\sqrt{4\rho - \lambda^2}}, 4\rho - \lambda^2 > 0. \quad (19)$$

As mentioned above, if the objective function about the error between α and α_a can be constructed, then $\alpha - \alpha_a$ can converge to zero within a finite time. According to (17), the objective function can be designed as

$$f(\alpha_a) = (\hat{F}_d)^2. \quad (20)$$

Based on (16), u_d can be expressed as

$$u_d = \frac{1}{\alpha_a} \left(-\omega_e i_q - \hat{F}_d - \beta_1 e_{rr} + \frac{d\hat{i}_d}{dt} \right). \quad (21)$$

According to (20) and (21), the gradient can be calculated as

$$\nabla f(\alpha_a) = 2\hat{F}_d \frac{d[(\alpha - \alpha_a)u_d]}{d\alpha_a} = -2\hat{F}_d u_d \approx \frac{-2p\hat{F}_d}{\alpha_a} \quad (22)$$

where p is the proportional coefficient, and its value determines the convergence speed. According to (21), the positive or negative value of p is related to i_q , and its size determines the speed of the convergence. If the p -value is larger, the convergence speed will be faster. However, an excessive p -value will lead to system instability. Therefore, selecting an appropriate p -value is necessary.

Then, applying the gradient of α_a to the FGM criterion, the formula obtained after discretization can be expressed as

$$\begin{cases} \alpha_a(k+1) = \alpha_a(k) - T_s \theta(k) \text{sgn}(\nabla f(\alpha_a(k))) \\ \theta(k+1) = (1 - T_s \lambda) \theta(k) + T_s \rho \|\nabla f(\alpha_a(k))\|. \end{cases} \quad (23)$$

The formula (16) after forward Euler discretization is expressed as

$$\begin{cases} e_{rr}(k) = \hat{i}_{dq}(k) - i_{dq}(k) \\ \hat{i}_{dq}(k+1) = \hat{i}_{dq}(k) \\ + T_s (D i_{dq}(k) + \hat{F}_{dq}(k) + \alpha_a(k) u_{dq}(k) + \beta_1 e_{rr}(k)) \\ \hat{F}_{dq}(k+1) = \hat{F}_{dq}(k) + T_s \beta_2 e_{rr}(k). \end{cases} \quad (24)$$

Based on the above AESO structure, the accurate prediction current and estimated disturbance can be obtained.

C. Reference Voltage Vector Calculation

Assuming $i_{dq}(k+1)$ reaches its reference value i_{dq}^{ref} , and considering the one-step delay in the digital control system, the voltage reference can be obtained

$$u_{dq}^{\text{ref}}(k+1) = \frac{i_{dq}^{\text{ref}} - \hat{i}_{dq}(k+1)}{\alpha_a(k+1)T_s} - \frac{\hat{F}_{dq}(k+1) + D\hat{i}_{dq}(k+1)}{\alpha_a(k+1)} \quad (25)$$

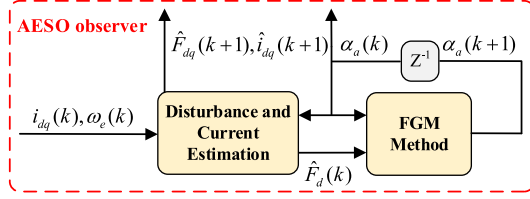


Fig. 2. Structure of the AESO.

where u_{dq}^{ref} is the reference value of the output voltage vector.

Fig. 2 shows the structure of the proposed AESO. First, $i_{dq}(k)$, $\omega_e(k)$ are sampled from the motor and input into the observer. Then, based on the observed disturbance and FGM criterion, the controller gain is adaptively updated. Finally, the updated controller gain is applied to the reference voltage prediction equation, and thus, u_{dq}^{ref} is calculated.

IV. PRINCIPLE AND DESIGN OF HARMONIC SUPPRESSION

A. Current Harmonic Analysis of PMSM

The error voltage in the synchronous coordinate system can be expressed as [34]

$$\begin{cases} V_{DT_d} = \frac{4V_{deadtime}}{\pi} \left[-\sin(\delta) - \sum_{n=6k}^{\infty} \left[\frac{\sin(n\omega_e t - \delta)}{n-1} + \frac{\sin(n\omega_e t + \delta)}{n+1} \right] \right] \\ V_{DT_q} = \frac{4V_{deadtime}}{\pi} \left[\cos(\delta) - \sum_{n=6k}^{\infty} \left[\frac{\cos(n\omega_e t - \delta)}{n-1} - \frac{\cos(n\omega_e t + \delta)}{n+1} \right] \right] \end{cases} \quad (26)$$

$k = 1, 2, 3, \dots$

where δ is the angle between the stator current vector and the q -axis.

The voltage distortion caused by VSI nonlinearity is the sixth harmonic and its multiples. Since the amplitudes of the remaining harmonics are negligible compared to the sixth harmonics, only the sixth harmonics are considered in this article. The voltage distortion $V_{DT_d_h6}$ and $V_{DT_q_h6}$ caused by VSI nonlinearity can be obtained as

$$\begin{cases} V_{DT_d_h6} = -\frac{4V_{deadtime}}{\pi} \left[\frac{\sin(6\omega_e t - \delta)}{5} + \frac{\sin(6\omega_e t + \delta)}{7} \right] \\ V_{DT_q_h6} = -\frac{4V_{deadtime}}{\pi} \left[\frac{\cos(6\omega_e t - \delta)}{5} - \frac{\cos(6\omega_e t + \delta)}{7} \right] \end{cases} \quad (27)$$

Under $i_d = 0$ control, the current vector completely falls on the q -axis, and the angle δ between it and the q -axis is zero. Then, the voltage distortion $V_{DT_d_h6}$ and $V_{DT_q_h6}$ caused by the nonlinearity of the inverter can be expressed as

$$\begin{cases} V_{DT_d_h6} = -\frac{48V_{deadtime}}{35\pi} \sin(6\omega_e t) \\ V_{DT_q_h6} = -\frac{8V_{deadtime}}{35\pi} \cos(6\omega_e t) \end{cases} \quad (28)$$

where $V_{deadtime} = (T_{deadtime}/T_p)u_{DC}$, $T_{deadtime}$ is a preset dead time to prevent short circuits in an inverter branch, T_p is the PWM switch period, and u_{DC} is the DC voltage.

Apart from the voltage distortion caused by the nonlinearity of the inverter, it is well known that the EMF harmonics

will also generate voltage distortion. By calculation, the back-EMF harmonic distortion in the d - q coordinate system can be expressed as [36], [37]

$$\begin{cases} V_{EMF_d} = \omega_e [\psi_5 \cos(6\omega_e t + \theta_5) + \psi_7 \cos(6\omega_e t + \theta_7)] \\ V_{EMF_q} = \omega_e [-\psi_5 \sin(6\omega_e t + \theta_5) + \psi_7 \sin(6\omega_e t + \theta_7)] \end{cases} \quad (29)$$

where ψ_5 and ψ_7 are the harmonic amplitude of the fifth- and seventh-order flux linkages, respectively.

As shown in (29), the voltage distortion is primarily the sixth-order component, and the voltage distortion caused by EMF harmonics increases with the rotor speed.

In summary, both VSI nonlinearity and EMF harmonics will generate disturbances. The distortion caused by EMF harmonics increases with the motor speed, and the deadtime effect is related to the PWM switching period. The overall voltage distortion can be expressed as

$$\begin{cases} V_{h_d} = V_{DT_d_h6} + V_{EMF_d} \\ V_{h_q} = V_{DT_q_h6} + V_{EMF_q} \end{cases} \quad (30)$$

According to the previous description, voltage distortion includes sixth-order harmonics caused by EMF harmonics and VSI nonlinearity. This article considers these two factors that cause distortion as a whole and designs a harmonic current controller that effectively suppresses current harmonics by adding a feedforward compensation voltage V_c to the synchronous reference frame.

The feedforward compensation voltage is as follows:

$$\begin{cases} V_{c_d} = w_{d6} \cos(6\omega_e t + \theta_{d6}) \\ V_{c_q} = w_{q6} \cos(6\omega_e t + \theta_{q6}) \end{cases} \quad (31)$$

where w_i is the amplitude of the compensation voltage, and θ_i is the phase of the compensation voltage.

B. Proposed Current Harmonics Suppression Method

This article adopts a compensation voltage generation method based on an adaptive neural network, and its mathematical model can be expressed as [38]

$$Y_n(x_i, w_i) = V_{c_dq} = \sum_{i=1}^n x_i(k)(w_i(k))^T = x(k)(w(k))^T \quad (32)$$

where $x(k) = [x_1(k), x_2(k)] = [\cos 6\omega_e kT_s, \sin 6\omega_e kT_s]$, $w(k) = [w_1(k), w_2(k)]$ are the input vector and corresponding weight factors, respectively, the output Y_n of a neural network is a linear function

To find the appropriate weight factor $w(k)$, define the error between V_{c_dq} and V_{h_dq} . Then, the least mean square algorithm is used in the neural network for adaptive learning of the weight factors, the goal is to minimize the defined error $e(k)$, so that each vector in the network is close to the learning object. The commonly used minimization objective function J can be expressed as

$$J = \frac{1}{2}|e(k)|^2 = \frac{1}{2}|V_{h_dq} - x(k)(w(k))^T|^2. \quad (33)$$

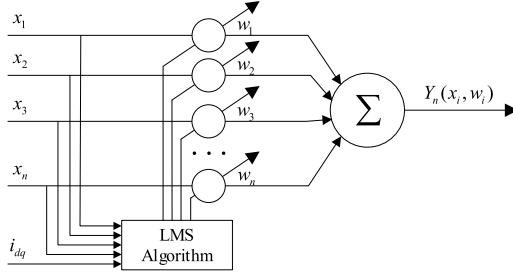


Fig. 3. Structure of the adaptive neural network.

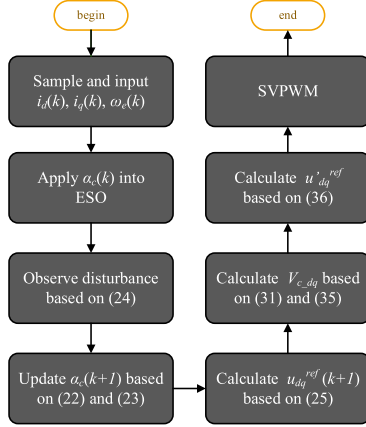


Fig. 4. Flow chart of the proposed method.

According to the gradient descent, the weight factor update formula is obtained

$$w(k+1) = w(k) - \mu_0 [\nabla J(w(k))] = w(k) + \mu_0 e(k)x(k) \quad (34)$$

where μ_0 is the step size that controls the convergence speed, the size of its value has a significant impact on the control system. A too-high value will cause the divergence of the system, while a too-low value will reduce the convergence speed.

In steady state, the motor resistance and inductance are considered constant values, so the error $e(k)$ between V_{c_dq} and V_{h_dq} can be replaced by the harmonic current i_{h_dq} as

$$e(k) = V_{h_dq} - V_{c_dq} = -Z_n i_{h_dq} \quad (35)$$

where the impedance modulus $Z_n = |R + jn\omega_e L_s|$.

Since the goal of adaptive learning is only to minimize the objective function J without the need to accurately calculate the value of $e(k)$, $e(k)$ can also be replaced by i_{dq} , which does not affect the direction of the gradient calculation. The final weight factor update formula can be expressed as

$$w(k+1) = w(k) - \mu_1 i_{dq}(k)x(k). \quad (36)$$

Compensating the calculated V_c into the reference voltage, the final reference voltage can be expressed as

$$u'_{dq}{}^{\text{ref}} = u_{dq}^{\text{ref}} + V_{c_dq}. \quad (37)$$

The structure of the adaptive neural network is shown in Fig. 3, the flow chart of the proposed method is shown in Fig. 4,

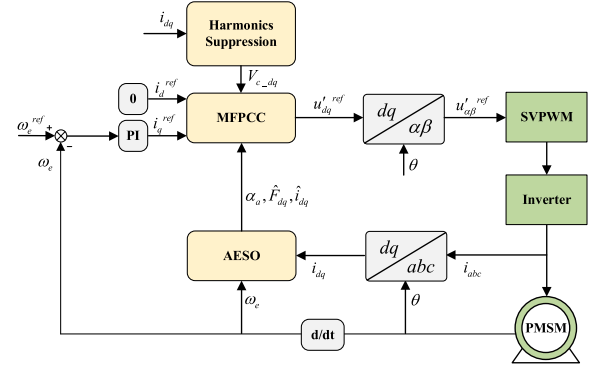


Fig. 5. Control diagram of the PMSM drive system based on AESO observer and harmonics suppression.

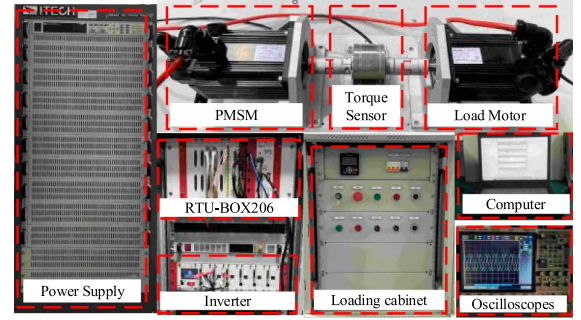


Fig. 6. Experimental platform of SPMSM drive.

TABLE I
MOTOR PARAMETERS

Parameter	Symbol	Value
Number of pole pairs	n_p	4
Stator inductance	L_s	0.84 mH
Stator resistance	R_s	0.245 Ω
Permanent-magnet flux linkage	ψ_f	0.13 Wb
DC-link voltage	U_{DC}	200 V
Rated current	$I_{\max}$	13.5 A
Rated speed	n	2500 rpm
Rated torque	T_e	15 N·m
Rotor inertia	J	0.0028 kgm ²

and the control diagram of the PMSM drive system based on the proposed method is shown in Fig. 5. Since the adaptive update of the weight factor does not depend on any motor parameters, this harmonic compensation method will not affect the parameter robustness performance of MFPCC.

V. EXPERIMENTAL RESULTS

To verify the effectiveness of the proposed method, a comparison was made between FOC (proposed in [6]), traditional MFPCC-ESO (proposed in [29]), MFPCC-ESO-IA (proposed in [39]), and MFPCC-AESO, considering harmonic compensation (HC-MFPCC-AESO), in the experimental results. The experimental setup is shown in Fig. 6, and the related PMSM parameters used in the experiment are listed in Table I. The

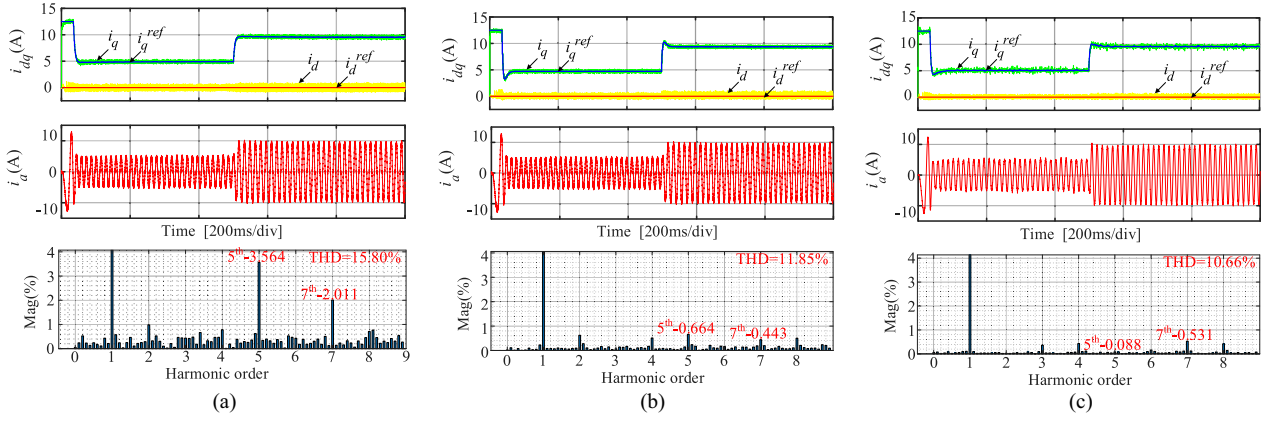


Fig. 7. Comparison of phase current, d- and q-axis current, and harmonic analysis under variable load conditions. (a) FOC. (b) Traditional MFPCC-ESO. (c) HC-MFPCC-AESO.

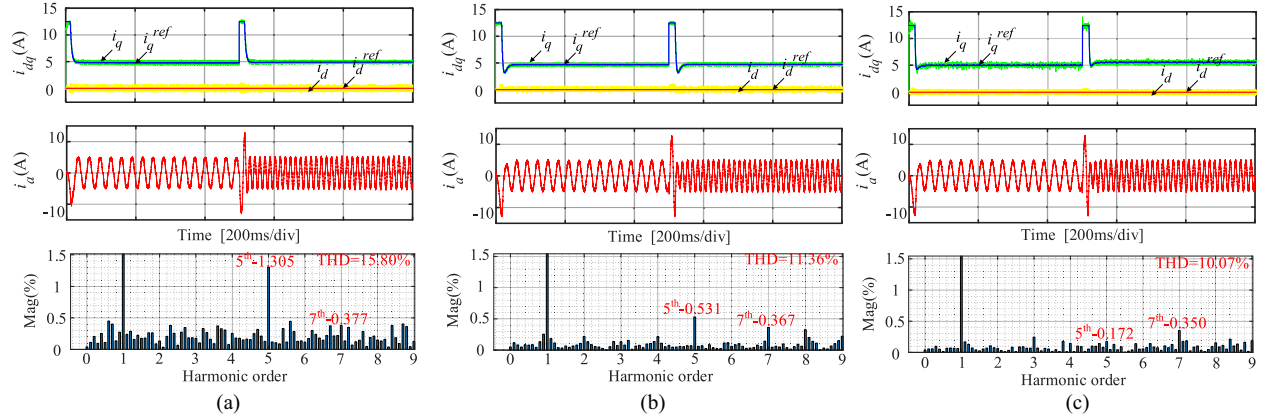


Fig. 8. Comparison of phase current, d- and q-axis current, and harmonic analysis under variable speed conditions. (a) FOC. (b) Traditional MFPCC-ESO. (c) HC-MFPCC-AESO.

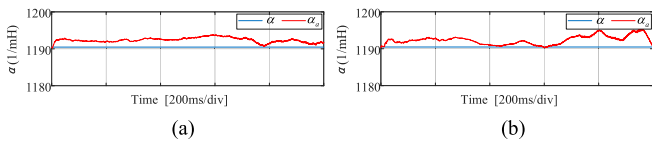


Fig. 9. Estimation of gain α_a in AESO. (a) Under variable load conditions. (b) Under variable speed conditions.

motor control experimental platform consists of a torque sensor, PMSM docking platform, RTUBOX-206 real-time digital controller using TMS320C28377 processor, loading cabinet, DC power supply, and two-stage VSI. The control frequency of the system is 10 kHz. To ensure fairness in comparison, the bandwidth and controller coefficients of ESO in traditional MFPCC are set to $\omega_b = 3000$ and $\alpha = 1190.5$, respectively. The parameter configuration for AESO in the proposed method is as follows: $\omega_b = 3000$, $\alpha = 1190.5$, $\lambda = 1e4$, $\rho = 2e8$, and the value of p is 10. In the harmonic suppression module, parameter μ_1 is set to $1e-5$. Moreover, in this experiment, all control methods employed the same speed loop PI controller,

with the parameters set as follows: proportional gain $k_p = 1.5$ and integral gain $k_i = 50$.

A. Experimental Results Under Precise Parameters

It is crucial to verify the transient-state control performance of the proposed control method in experiments. Figs. 7 and 8 show the results of FOC, traditional MFPCC-ESO, and the proposed HC-MFPCC-AESO under precise motor parameters. The experimental conditions in Fig. 7 are that the speed reference value remains constant at 1000 rpm and the load torque changes from 5 to 10 N·m, while the experimental conditions in Fig. 8 are that the load torque remains constant at 5 N·m and the speed reference value varies from 500 to 1000 rpm. The three channels display the stator current of phase A, the reference current and actual current of the d- and q-axis, and the fast Fourier analysis of the phase current. Since under low-speed and light-load conditions, issues such as harmonic interference and torque pulsation in PMSM become more pronounced, this article selects such operating points to more clearly demonstrate the effectiveness of the proposed method in addressing these key control challenges. Moreover, it is important to note that the FFT analyses presented in both figures were performed under

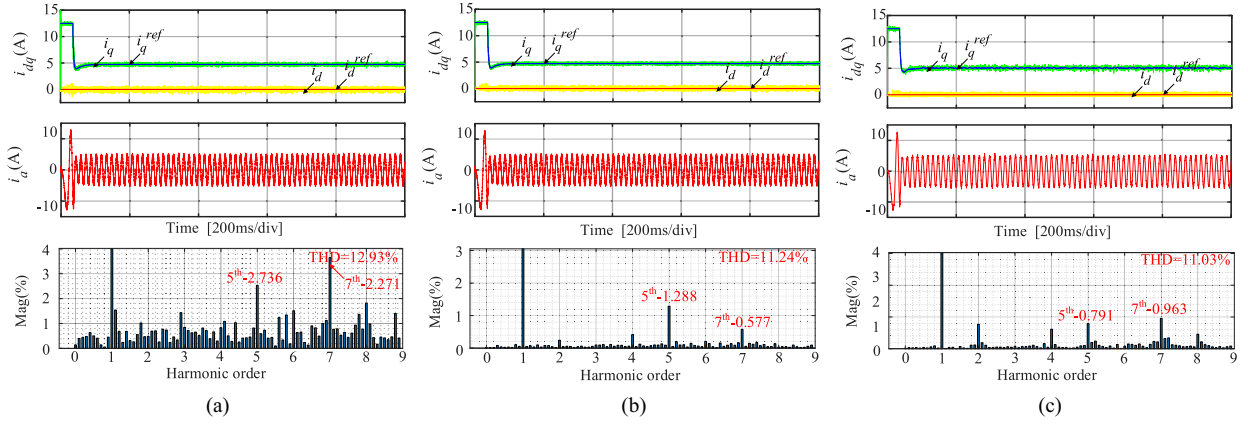


Fig. 10. Comparison of phase current, d - and q -axis current, and harmonic analysis under 200% L_s . (a) Traditional MFPCC-ESO. (b) MFPCC-ESO-IA. (c) MFPCC-AESO.

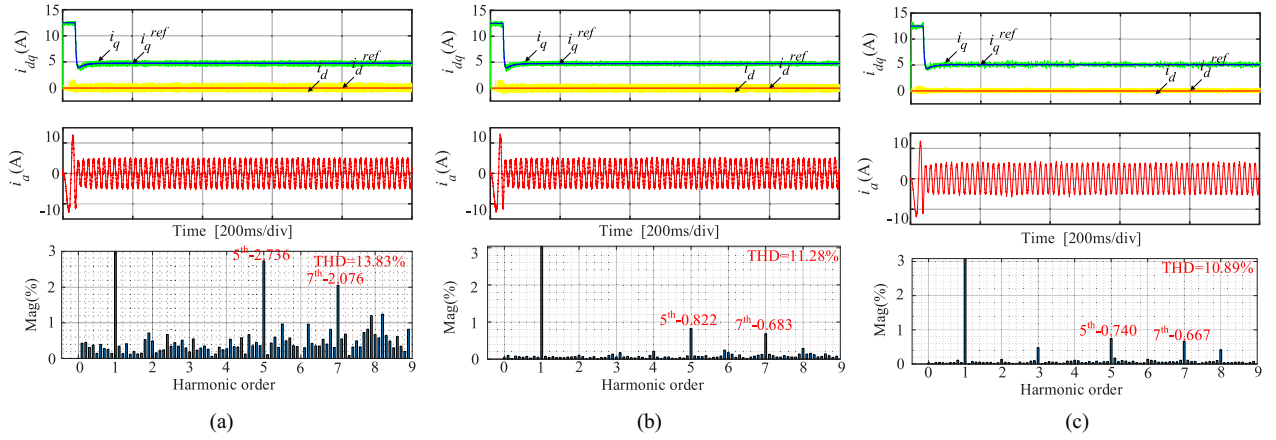


Fig. 11. Comparison of phase current, d - and q -axis current, and harmonic analysis under 50% L_s . (a) Traditional MFPCC-ESO. (b) MFPCC-ESO-IA. (c) MFPCC-AESO.

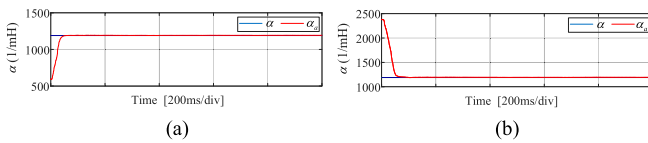


Fig. 12. Estimation of gain α_d in AESO. (a) Under 200% L_s . (b) Under 50% L_s .

the steady-state condition of 1000 rpm and 5 N·m. It can be clearly seen that the motor can run smoothly and efficiently under precise motor parameters regardless of speed and load mutation; however, MFPCC-ESO has a smaller current ripple than FOC, and the proposed HC-MFPCC-AESO method further reduces the current ripple, and as can be seen from the fast Fourier analysis figure that the harmonic compensation method effectively reduces harmonic currents. Under two different operating conditions, the total harmonic distortion (THD) decreased from 11.85% and 11.36% to 10.66% and 10.07%, respectively. In addition, as shown in Fig. 9, the α_c value in the controller can track the actual value of the reciprocal inductance very well, with only some small fluctuations to a certain extent.

Therefore, it can be concluded that the performance of ESO and AESO is nearly identical under precise parameters, and the reduction in THD observed in the experimental results is attributable to the performance improvement achieved by the harmonic current control method. In conclusion, the above results show that the proposed HC-MFPCC-ESO exhibits superior control performance under transient disturbances. Moreover, it is expected that even under rated load or rated speed conditions, the AESO can still effectively suppress the impact caused by parameter variations, ensuring the q -axis current tracks the reference value with high precision. Compared with the traditional MFPCC-ESO method, the steady-state error will be further reduced, and the THD of the stator current can be maintained at a lower level.

B. Experimental Results Under Parameters Mismatch

It is difficult to change the motor parameters in the experiment. Therefore, in order to verify the effectiveness of the proposed method in suppressing parameter mismatch disturbances, the article verifies the control performance of the three methods in the case of parameter mismatch by changing the parameters

TABLE II
IMPLEMENTATION COMPLEXITY OF FOUR METHODS

	FOC	Traditional MFPCC-ESO	HC-MFPCC-AESO
Required motor parameters	R_s, L_s, Ψ_f , etc.	L_s	None
Parameters to be designed	None	α, ω_b	$\alpha, \omega_b, \lambda, \rho, p, \mu_1$

in the control program. Figs. 10 and 11 show the results of the three methods under parameter mismatch. It is worth noting that, to ensure the fairness of the experiment, we disabled the harmonic compensation strategy in the proposed method. The test condition is that the reference speed value and load torque are kept at 1000 rpm and 5 N·m, while two tests are conducted with the L_s in the controller set to 50% of the actual machine value in one test and 200% of the actual machine value in the other. As shown in the figure, parameter mismatch has a significant impact on the traditional MFPCC. When the inductance value is severely mismatched, the estimation performance of the traditional MFPCC method is not satisfactory, and the q -axis current cannot accurately track the reference value and presents a slight deviation. This result highlights the significant impact of inductance on controller performance. In contrast, the steady-state tracking error of q -axis current in MFPCC-ESO-IA and MFPCC-EASO does not change significantly. As shown in Fig. 12, the estimated value of controller parameter α_c clearly indicates that AESO significantly improves anti-interference capability in the presence of parameter errors compared to the traditional ESO. This improvement can be attributed to the real-time dynamic adjustment of AESO controller gain based on FGM criteria. Therefore, even in the presence of severe inductance mismatches, AESO shows commendable control performance, showing strong robustness to machine parameter changes. At the same time, the THD of the three methods is 12.93%, 11.23%, and 11.03%, respectively, under the condition of 200% L_s . It can be observed that the MFPCC-ESO-IA method adaptively optimizes the controller parameters through the introduction of an immune algorithm, but its performance improvement is not as significant as that of the proposed method. This, from another perspective, demonstrates that the proposed method can adaptively update the controller gains, avoid disturbances caused by parameter mismatches, and achieve a better steady-state current waveform.

The complexity comparison of the FOC, traditional MFPCC-ESO, and HC-MFPCC-ESO is listed in Table II. Fig. 13 shows the execution times of different methods. It can be seen that the FOC method has the shortest calculation time. Moreover, due to the need for compensating for the feedforward decoupling term, FOC requires the input of basic motor parameters in the current loop. Compared with the previous two methods, the proposed method slightly increases the computation time, but due to the absence of any motor parameters, the robustness of the method to parameter mismatches is greatly enhanced. In

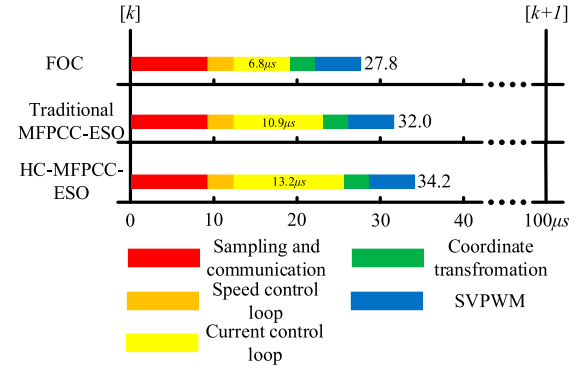


Fig. 13. Comparison of execution time with different methods.

addition, although the proposed method requires adjustment of relatively more controller parameters, it is still within an acceptable range. Considering its good overall performance, it can be concluded that the HC-MFPCC-ESO method is effective.

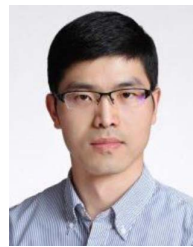
VI. CONCLUSION

In this article, an improved MFPPC method is proposed, and harmonic suppression is considered to improve the steady-state performance and parameter robustness of motor control systems. First, based on the ultralocal model, an AESO-based MFPCC method is proposed, which adaptively updates the controller gain by combining FGM criteria, effectively eliminating the impact of parameter mismatch. Meanwhile, the proposed harmonic current control method can effectively control low-order current harmonics. The experimental results confirm that the proposed method has excellent performance in current ripple, tracking accuracy, and achieves good parameter robustness.

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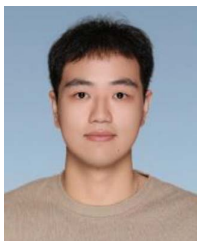
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